

## Logistic Regression:

### Introduction

Logistic Regression is a supervised machine learning algorithm used for classification problems, especially when the target variable is categorical. Despite its name, logistic regression is used for classification rather than regression. It predicts the probability that an instance belongs to a particular class.

For example:

- Predict whether an email is Spam (1) or Not Spam (0).
- Predict whether a patient has Diabetes (Yes/No).
- Predict whether a customer will Buy a Product (Yes/No).

### Definition

Logistic Regression is a statistical model that estimates the probability of a binary outcome using the logistic (sigmoid) function. It converts a linear combination of input variables into a probability value between 0 and 1.

### Why Not Linear Regression?

Linear regression predicts continuous values that can be less than 0 or greater than 1.

Example:

If Linear Regression predicts:

$$P(Y = 1) = 1.5$$

or

$$P(Y = 1) = -0.3$$

These are impossible probabilities.

Logistic Regression solves this problem by using the **Sigmoid Function**, which always produces values between 0 and 1.

### Logistic (Sigmoid) Function

The sigmoid function is

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

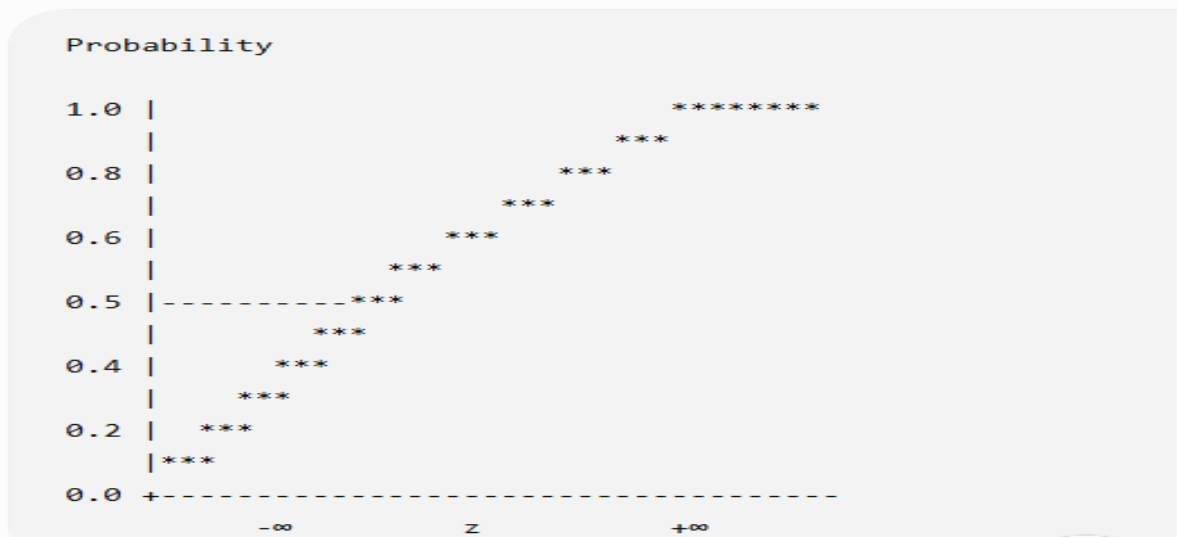
where

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where

- $P(Y = 1)$  = Probability of positive class
- $e$  = Euler's constant ( $\approx 2.718$ )
- $\beta_0$  = Intercept
- $\beta_1, \beta_2, \dots, \beta_n$  = Model coefficients
- $x_1, x_2, \dots, x_n$  = Independent variables

# Sigmoid Curve



## Working of Logistic Regression

### Step 1: Input Features

Example:

Age = 40

Income = 60000

### Step 2: Linear Equation

Compute

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Suppose

$$z = 2.3$$

### Step 3: Apply Sigmoid

$$P = \frac{1}{1 + e^{-2.3}}$$

$$P = 0.909$$

Meaning there is a **90.9% probability** of belonging to Class 1.

### Step 4: Decision Boundary

If

Probability  $\geq 0.5$

Predict Class = 1

Else

Predict Class = 0

## Types of Logistic Regression

### 1. Binary Logistic Regression

Only two classes.

Examples

- Pass / Fail
- Yes / No
- Male / Female
- Spam / Not Spam

### 2. Multinomial Logistic Regression

More than two classes.

Examples

- Red
- Blue
- Green

### 3. Ordinal Logistic Regression

Output classes have an order.

Examples

- Low
- Medium
- High

## Cost Function

Unlike Linear Regression, Logistic Regression uses **Log Loss (Binary Cross-Entropy)**.

For one observation,

$$\text{Loss} = -(y \log(p) + (1 - y) \log(1 - p))$$

where

- $y$  = Actual class
- $p$  = Predicted probability

The objective is to minimize this loss during training.

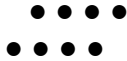
## Training Process

1. Initialize coefficients.
2. Calculate predicted probabilities.
3. Compute the loss.
4. Update coefficients using optimization algorithms such as Gradient Descent.
5. Repeat until convergence.

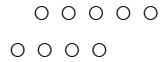
## Decision Boundary

The model separates classes using a decision boundary.

Example:      Class 1



----- Decision Boundary



Class 0

### **Assumptions of Logistic Regression**

1. Dependent variable is categorical.
2. Observations are independent.
3. Little or no multicollinearity among independent variables.
4. Linear relationship between predictors and the log-odds of the outcome.
5. Large sample size is preferred.

### **Advantages**

- Simple and easy to implement.
- Fast training.
- Produces probabilities.
- Works well for binary classification.
- Easy to interpret model coefficients.
- Less prone to overfitting than many complex models.

### **Disadvantages**

- Cannot capture complex non-linear relationships without feature engineering.
- Sensitive to outliers.
- Performance decreases with highly correlated features.
- Assumes a linear relationship between predictors and log-odds.

### **Applications**

- Disease diagnosis
- Credit scoring
- Fraud detection
- Customer churn prediction
- Spam email detection
- Sentiment analysis
- Loan approval
- Marketing response prediction

```
[1]: from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Load dataset
data = load_breast_cancer()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

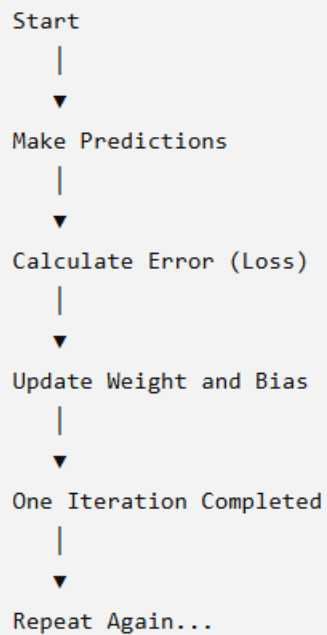
# Create model
model = LogisticRegression(max_iter=1000)

# Train model
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))
```

## Visual Representation



## What is a Weight?

A weight tells the model how important a particular input feature is in making the prediction.

- A large positive weight means the feature strongly increases the probability of the positive class.
- A large negative weight means the feature strongly decreases the probability.
- A weight close to zero means the feature has little influence.

### Example

Suppose we want to predict whether a student will pass.

| Feature        | Value |
|----------------|-------|
| Study Hours    | 8     |
| Attendance (%) | 90    |

Assume the model has learned:

- Weight for Study Hours = 0.8
- Weight for Attendance = 0.2

The equation becomes:

$$z = 0.8 \times \text{Study Hours} + 0.2 \times \text{Attendance} + b$$

Here, **Study Hours** has a larger weight, so it has a greater effect on the prediction than **Attendance**.

## What is Bias?

The **bias** is a constant value added to the equation.

It allows the model to make predictions even when all input features are zero and helps shift the decision boundary to fit the data better.

### Example

Suppose:

- Study Hours = 0
- Attendance = 0

Without bias:

$$z = 0$$

With bias:

$$z = 0 + (-3) = -3$$

The bias changes the predicted probability, allowing the model to better represent the data.

### Problem Statement

Predict whether a student will **Pass (1)** or **Fail (0)** based on the number of study hours.

### Dataset

| Student | Study Hours (x) | Actual Result (y) |
|---------|-----------------|-------------------|
| S1      | 1               | 0 (Fail)          |
| S2      | 2               | 0 (Fail)          |
| S3      | 3               | 0 (Fail)          |
| S4      | 4               | 1 (Pass)          |
| S5      | 5               | 1 (Pass)          |

Assume the Logistic Regression model has already learned the following parameters:

- Weight (w) = 1
- Bias (b) = -3

### Step 1: Calculate the Linear Equation

The logistic regression equation is

$$z = wx + b$$

Since

$$w = 1, \quad b = -3$$

then

$$z = x - 3$$

## Step 2: Calculate Probability Using Sigmoid Function

The sigmoid function is

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

### Instance 1

Study Hours = 1

Calculate z

$$z = (1 \times 1) - 3 = -2$$

Probability

$$P = \frac{1}{1 + e^2}$$

Since

$$e^2 = 7.389$$

$$P = \frac{1}{8.389} = 0.119$$

Probability = 11.9%

Prediction:

Since  $0.119 < 0.5$

Predicted Class = 0 (Fail)



## Instance 2

Study Hours = 2

### Calculate z

$$z = 2 - 3 = -1$$

### Probability

$$P = \frac{1}{1 + e^1}$$

$$P = \frac{1}{3.718} = 0.269$$

Probability = 26.9%

Prediction:

Fail (0)

## Instance 3

Study Hours = 3

### Calculate z

$$z = 3 - 3 = 0$$

### Probability

$$P = \frac{1}{1 + e^0}$$

$$P = \frac{1}{2} = 0.50$$

Probability = 50%

Prediction:

Since Probability  $\geq 0.5$

## Final Results

| Student | Study Hours | Actual Class | $z = x - 3$ | Probability | Predicted Class |
|---------|-------------|--------------|-------------|-------------|-----------------|
| S1      | 1           | 0            | -2          | 0.119       | 0 (Fail)        |
| S2      | 2           | 0            | -1          | 0.269       | 0 (Fail)        |
| S3      | 3           | 0            | 0           | 0.500       | 1 (Pass)        |
| S4      | 4           | 1            | 1           | 0.731       | 1 (Pass)        |
| S5      | 5           | 1            | 2           | 0.881       | 1 (Pass)        |

## Accuracy Calculation

Compare the actual and predicted classes:

| Student | Actual | Predicted | Correct? |
|---------|--------|-----------|----------|
| S1      | 0      | 0         | ✓        |
| S2      | 0      | 0         | ✓        |
| S3      | 0      | 1         | X        |
| S4      | 1      | 1         | ✓        |
| S5      | 1      | 1         | ✓        |

Number of correct predictions = 4

Total instances = 5

$$\text{Accuracy} = \frac{4}{5} \times 100 = 80\%$$